

Phase 6

Project Testing Phase

API test cases, results and evidence

1. Testing Objective and Approach

The goal is to verify that the backend REST APIs work end to end: registering and logging in, uploading study material and generating AI resources through the Google Gemini API. Testing was performed manually with **Postman** against the server running locally on `http://localhost:5000`. Protected AI endpoints were called with the JWT access token obtained at login.

2. Test Environment

Item	Detail
Server	Node.js / Express on localhost:5000 (npm start)
Database	MongoDB (connected before server start)
API client	Postman
AI model	Google Gemini 2.5 Flash
Test material	Node.js Introduction.txt (title: Node.Js)

3. Test Cases and Results

ID	Feature	Request	Expected	Result (from Postman)
TC-01	User registration	POST /api/auth/register body: name, email, password, role	201 Created; success message, tokens and user details returned	Pass - 201 Created, 200 ms
TC-02	User login	POST /api/auth/login body: email, password	200 OK; access and refresh tokens returned	Pass - 200 OK, 154 ms
TC-03	Upload study material	POST /api/materials/upload form-data: file, title	201 Created; 'Material uploaded' with stored content	Pass - 201 Created, 124 ms
TC-04	AI summary	POST /api/materials/:id/summarize	200 OK; concise summary returned	Endpoint documented; no screenshot in source document
TC-05	AI flashcards	POST /api/materials/:id/flashcards	200 OK; list of question/answer flashcards	Pass - 200 OK, 5.49 s
TC-06	AI quiz	POST /api/materials/:id/quiz	200 OK; MCQs with options and answer	Pass - 200 OK, 7.34 s
TC-07	AI study plan	POST /api/materials/:id/study-plan body: goal, hoursPerDay, days	200 OK; personalized study plan text	Pass - 200 OK, 19.65 s

Test data used

- Registration / login: name 'admin', email admin@example.com, role 'admin'.
- Study plan request: goal 'Prepare for my backend development exam', hoursPerDay 2, days 7.

TC-01: User registration



Figure 2: POST /api/auth/login - 200 OK

AI StudyBuddy | Node.js | Express.js | MongoDB | Google Gemini AI

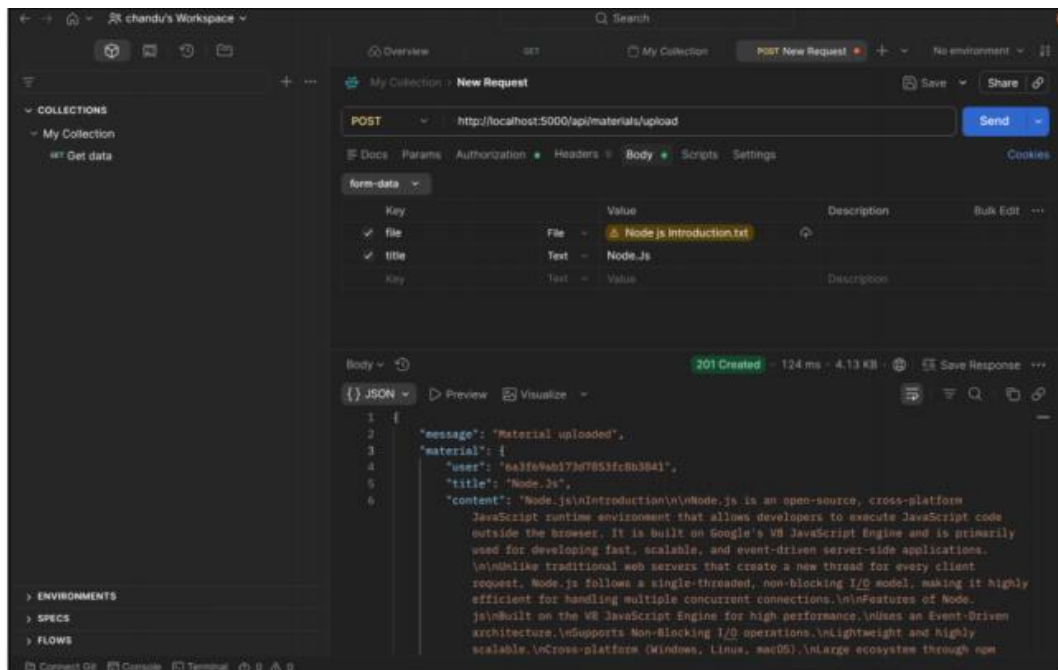


Figure 3: POST /api/materials/upload - 201 Created

TC-05: AI flashcards

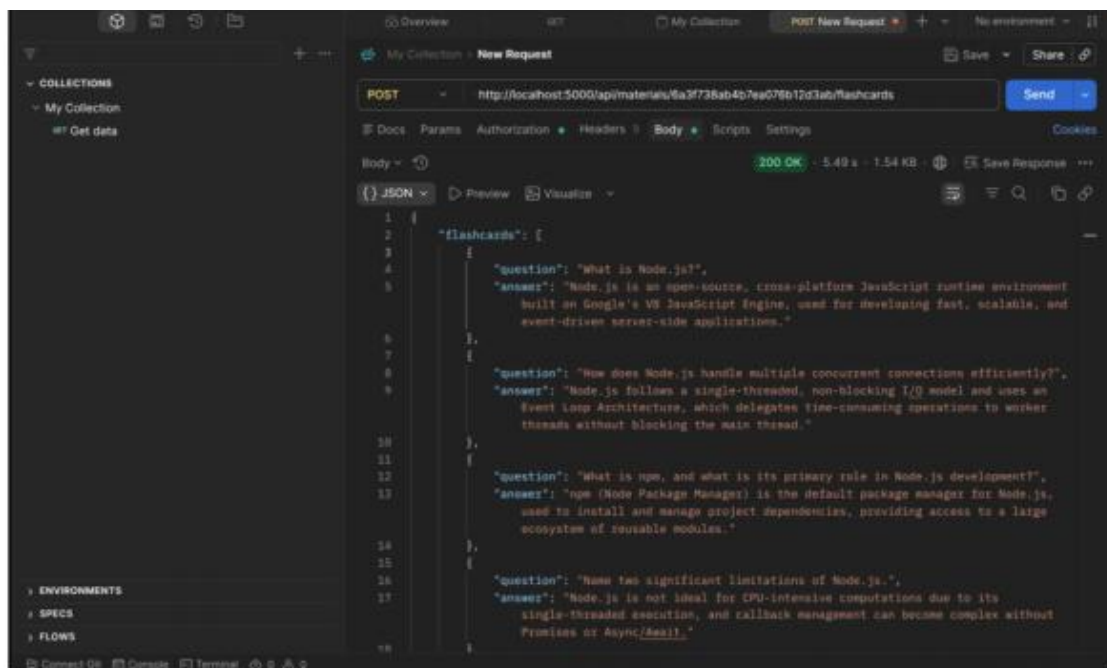


Figure 4: Flashcards generation - 200 OK

TC-06: AI quiz

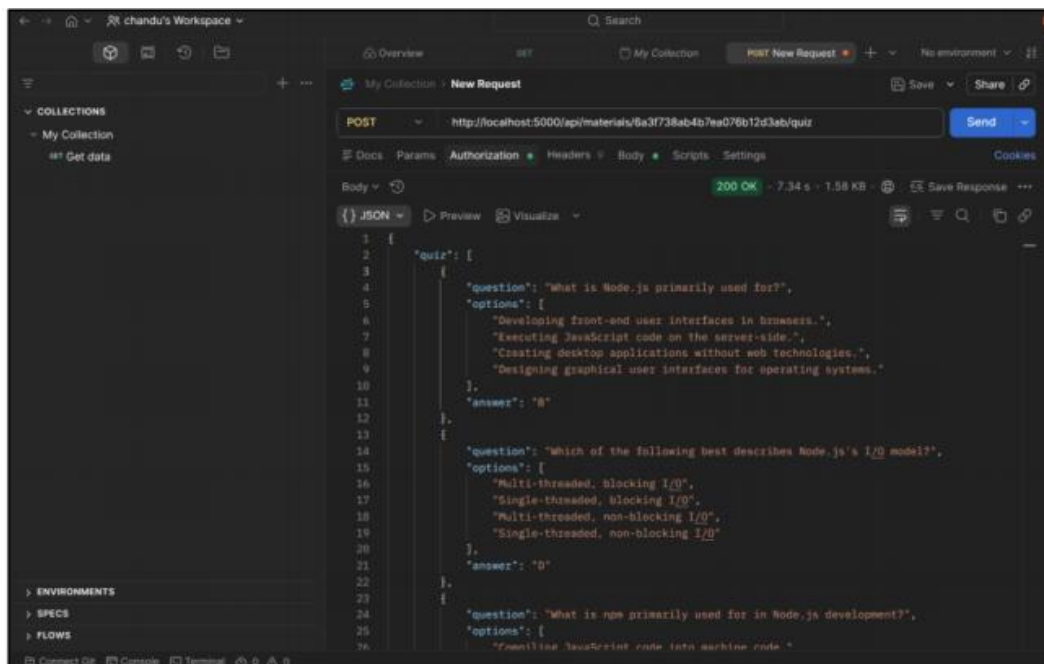


Figure 5: Quiz generation - 200 OK

TC-07: AI study plan

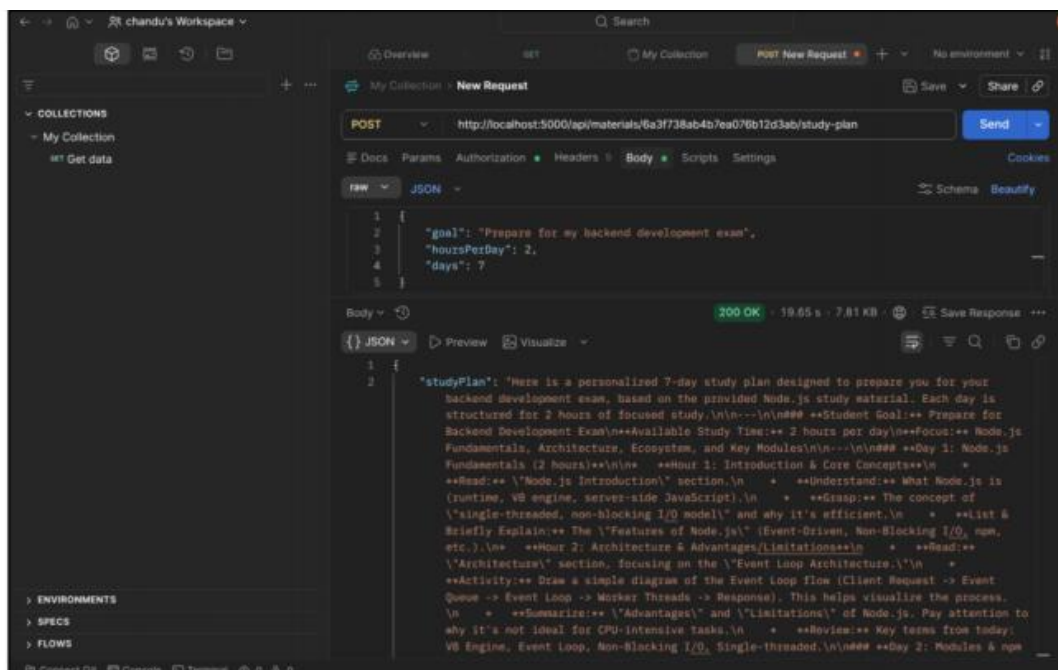


Figure 6: Study plan generation - 200 OK

5. Observations

- Authentication and upload endpoints respond in about 120 to 200 ms.
- AI endpoints take longer because they wait on the Gemini API: flashcards about 5.5 s, quiz about 7.3 s, study plan about 19.7 s. Frontends should show a loading indicator.
- Flashcard and quiz responses are structured JSON (question, options, answer) that map directly to the Material schema.
- Password hashing, the 401 (not authenticated / invalid token) and 403 (admins only) checks and the file-type / 5 MB upload limits are implemented in the code; the source document does not include tests that trigger them, and the summary and admin endpoints were not shown being tested. These are good candidates for the next round of testing.